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Brian Brown

https://github.com/brian-brow
US Citizen

Education

University of Florida — Gainesville, FL
B.S. in Computer Science, GPA: 3.68/4.0 — *Jun 2022 – May 2026*

- **Coursework**: Data Structures & Algorithms, Operating Systems, Algorithm Abstraction & Design, Computer Organization, Machine Learning Engineering, Natural Language Processing

Work Experience

Loxahatchee River District — Jupiter, FL
IT Intern — *May 2025 - Present*

- Designed and consumed RESTful APIs to automate ingestion and storage of district call and training data.
- Built automated extraction pipelines for water-distribution data, reducing manual collection time by several hours per week.
- Developed a database-driven tool to auto-generate late-notice letters for customers, improving turnaround time for communications.
- Created a Linux-based monitoring utility that packages Kismet to monitor district pump stations, leveraging command-line tools and system administration skills.
- Assisted in implementing AI-based image recognition for pressure gauge readings and streamlining recording assets into an asset management database.
- Worked with database-backed services and internal tools, with exposure to C#/.NET-based systems.

Kratos Defense & Solutions — Jupiter, FL
Digital Engineering Intern — *May 2023 – Aug 2023*

- Designed functional models in Cameo Systems Modeler to calculate the optimal inputs for a turbojet.
- Wrote Visual Basic scripts in Excel to calculate the expected thrust and cost of a hypothetical engine based on size.
- Collaborated with engineering teams to implement performance calculations for a C++ steady-state engine model.

Projects

- **CEN3031 Group Project: "Craven"**: Designed and implemented a web application enabling users to browse and match with recipes based on preferences, featuring a suggestion algorithm. Built RESTful backend services using Node.js and Express for authentication and user data. Designed MongoDB schemas to store user preferences and saved recipes and integrated an external recipe API. Developed a React.js frontend consuming backend APIs.
- **Boid Flocking Simulation**: Implemented a real-time Boids flocking simulation with configurable rule weights, using Node.js/Express backend and a React + p5.js frontend.
  - Link to website hosting simulation: https://boid-simulation.web.app/
- **3D Graphics Engine**: Designed a 3D Graphics Engine in OpenGL and SDL and created a traversable environment within it.
- **Arch Linux Custom Rice**: Built and maintained a customized Arch Linux daily-driver environment, including bash-based system utilities, a Qt todo-list application, and custom desktop widgets.

Involvement

- **University of Florida Symphonic Band**: 100-piece audition-based band led by Professor Jay Watkins, Associate Director of Bands
- **PMOC (Passionate Musicians On Campus)**: Organization dedicated to performing music at nursing or retirement homes, and teaching younger musicians the basics of their instruments.
- **UF Basketball Pep Band**: Audition-based pep band performing at all UF home basketball games.

Skills

Languages: Python, Java, JavaScript, C++, C, Bash
Technologies: Linux (CLI/shell scripting), Docker (project experience), RESTful APIs, SQL (PostgreSQL, MySQL, SQLite), Node.js, .NET / ASP.NET, React, Git, HTML/CSS, Qt